

Why SafeEvolve?

- Raw evolution optimizes utility.
- Unsafe successes can become persistent skills.
- Safety must govern both writing and reuse.

SafeEvolve: Governing Skill Admission and Reuse

A safety-aware pipeline for self-improving agent skill libraries

Core Decision Points

- 1 Write gate: admit, repair, or reject?
- 2 Reuse gate: retrieve, suppress, or retire?

This trajectory worked — should I turn it into a skill?



1 Trajectory Batch

Tasks + tool traces + observed success



2 Candidate Skill Bundle

Procedure + resources + native method output



Candidate may contain useful steps... and unsafe generalization.

3 Write-Time Governance

- Admission audit
 - Targeted repair
 - Verification preservation
- ✗ / ✎ / ✓ Reject / Repair / Admit

Goal: block unsafe generalization at write time.

Repair it, don't blindly save it.



Store only the repaired, verified, and traceable skill.

4 Governed Skill Library

Accepted skill + lineage



source revision
risk prior provenance

Reuse-Time Governance

- Safety-aware selection
 - Risk prior
 - Retrieval evidence
- ⊘ / ✓ Suppress / Retrieve

Goal: stop retained risk before execution.



Should this skill be retrieved for a later task?

6 Later Task Reuse

Agent action + outcome



✓ useful and safe
✗ unsafe reuse

I reused the skill on a new task.

7 Feedback / Governance Update

Rollback / quarantine / repair / retire



Observed reuse outcome Utility signal Safety signal

If reuse cause harm, quarantine or retire the skill.



Write-Time Checks



- Audit instructions
- Inspect attached resources
- Preserve verification steps
- Add provenance metadata



Useful is not enough — it must also be safe.

Library Metadata

- lineage
- source & revision history
- risk prior
- repair history



Trace where this skill came from.

Reuse-Time Evidence

- retrieval history
- observed safety outcomes
- utility outcomes
- retirement trigger



Evidence today prevents risk tomorrow.

Legend (at a glance)

- ✓ Safe outcome
- ✗ Unsafe outcome
- ↻ Feedback loop
- ✓ Governance gate
- ⚠ Risk / Caution
- 🔒 Persistent memory

SafeEvolve keeps useful procedures, but governs the persistent update boundary.